FINAL EXAM PROJECT – STUDENT CHECKLIST (A–Z)

Course: Software Deployment, Operations & Maintenance
Assessment: Final Exam Project (50%)

CRITICAL RULE

Any feature, improvement, or advanced functionality will be recognized for grading ONLY IF it:

- appears in the video demo and/or live in-class presentation, AND
- is clearly explained and evidenced in the technical report.

If a feature is implemented but not shown in the demo and not documented in the report, it will be treated as non-existent for grading purposes.

A. Architecture Selection

You must select one architecture tier and clearly state it in the technical report.

Tier 1: Single-Server, Non-Containerized Deployment
Tier 2: Single-Server, Containerized Deployment (Docker Compose)
Tier 3: Multi-Server Architecture with Centralized Load Balancing
Tier 4: Docker Swarm–Based Orchestration
Tier 5: Kubernetes-Based Architecture
The deployed system must actually match the selected tier in practice. The system must be operational and reachable.

B. Infrastructure & Environment

A real cloud provider is used (AWS, Azure, GCP, Oracle Cloud, DigitalOcean, etc.).
At least one Ubuntu server is provisioned and accessible.
A registered domain name points to the deployed system.
HTTPS is correctly configured using Let's Encrypt or an equivalent CA.

Provisioning approach:

- Manual provisioning (acceptable, lower score), or
- Infrastructure as Code (Terraform ± Ansible, higher score).

Evidence of infrastructure setup is included in the technical report.

C. CI/CD Pipeline (CORE REQUIREMENT)

Continuous Integration (CI)

The pipeline is triggered automatically by a code push.

CI includes:

- linting or code quality checks
- dependency caching
- build artifact generation
- security scanning (Trivy / Snyk / SonarQube / equivalent)

The pipeline fails on Critical/High vulnerabilities. For containerized tiers (Tier 2–5), Docker images are built, version-tagged, and pushed to a registry.

Continuous Delivery (CD)

CD automatically deploys to production:

- pull correct artifact/image version
- update deployment configuration
- deploy new version
- restart services safely

If staging and production are both used:

- environment separation
- manual approval gate
- documented deployment strategy

D. Deployment & Orchestration

Deployment behavior matches the selected tier.

Tier expectations:

- **Tier 1**: systemd or PM2, auto-restart after reboot
- **Tier 2**: Docker Compose, multi-container, persistent volumes
- **Tier 3**: multiple servers with managed cloud load balancer preferred
- **Tier 4**: Docker Swarm cluster with replicated services
- **Tier 5**: Kubernetes with Services, Ingress, self-healing Horizontal scaling is demonstrated where applicable.

E. Monitoring & Observability

Prometheus is installed and collecting metrics.
Grafana dashboards are accessible.

Dashboards show:

- CPU usage
- memory usage
- service/container/pod status

Monitoring reflects the real production system and is shown during the demo.

F. Mandatory Demo Scenario (VIDEO / LIVE)

- A real source code change is made, committed, and pushed.
- CI pipeline runs automatically and visibly.
- CD deploys the new version automatically.
- Production website shows the updated version.
- HTTPS validity is verified.
- Grafana dashboards are opened and explained.
- A failure is simulated and system behavior is observed.

G. Technical Report (PDF)

The report follows the official academic template and is written in formal academic English.

The report includes exactly five chapters:

1. Overview & System Architecture
2. Infrastructure Provisioning
3. CI/CD Pipeline Design
4. Deployment & Orchestration
5. Monitoring, Observability & Lessons Learned

Architecture diagrams, evidence, and design justifications are included.

H. Video Demonstration

The video follows the mandatory demo scenario step by step. CI/CD execution, deployment, monitoring, and failure behavior are clearly visible.

I. Submission Deliverables

A single .zip file is submitted containing:

- source code
- infrastructure files
- CI/CD pipeline configurations
- deployment manifests
- monitoring configurations
- technical report (PDF)

The production website URL is included and accessible via HTTPS. All artefacts describe the same system.

J. Final Self-Check

- The system is running via HTTPS.
- CI/CD works without manual intervention.
- All features appear in both the demo and the report.
- No placeholder or staged content is used.